



GRADUATION PROJECT

TinyRL for Anomaly Detection in Smart IoT-Assisted Factories

Team:

Hamza Hussein Alaa Al-Deen, Abdullah Nidal Samarah, Mohammad Jaafar Jibreel, Ammar Hmead, Wael Dergham

University:

Al Zaytoonah University of Jordan

Supervisor:

Dr. Yousef Shnaiwer

Introduction

Why AI-Based Motor Monitoring?

INDUSTRY PROBLEM

~45%

of industrial downtime
caused by motor failures

\$50B+

annual losses from
unexpected equipment failure

70%

of failures preventable
with early detection

OUR APPROACH



Sound-Based Monitoring

Capture motor audio with MEMS microphone for non-invasive real-time analysis



AI Classification

CNN deep learning model trained on MIMII industrial dataset



IoT Integration

ESP32 WiFi module for wireless data transmission to cloud server



Instant Alerts

Immediate operator notification via LEDs, buzzer, and dashboard

Project Objectives

01

Real-Time Monitoring



Continuous processing of ~~dataset~~ audio files simulating ~~motor~~ sounds at 8000 Hz for real-time anomaly classification

02

Early Fault Detection



Identify anomalies before mechanical failure occurs, reducing unplanned downtime

03

Low-Cost Implementation



Use affordable ESP32 + MEMS microphone instead of expensive industrial sensors (< \$20 ~~BOM~~)

04

AI-Based Classification



CNN model with spectrogram features achieving high accuracy on normal vs abnormal classes

05

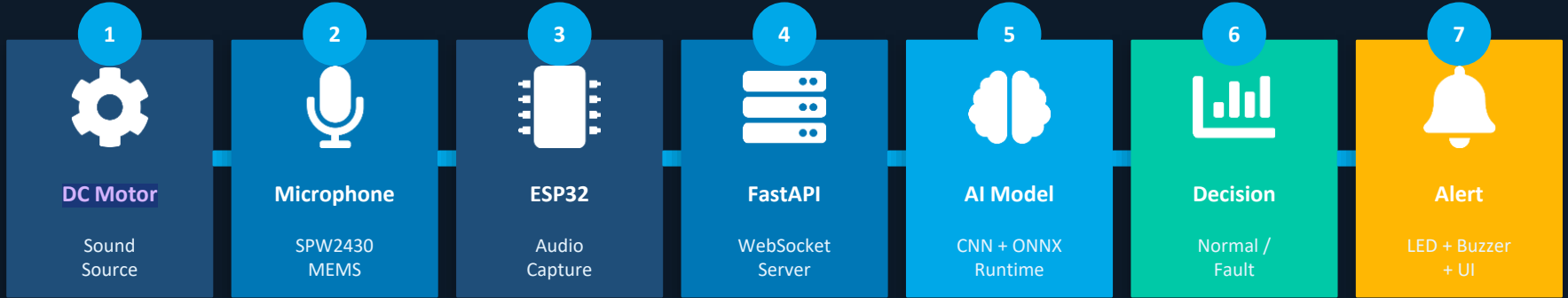
Remote IoT Monitoring



WiFi-enabled ~~WebSocket~~ streaming for real-time remote access from any browser

System Overview

End-to-end data flow from motor to decision



System Description

Note: In our implementation, machine audio is simulated using pre-recorded sounds from the MIMII dataset — no physical microphone acquisition was performed. The dataset audio files are fed directly into the FastAPI server, which preprocesses them into mel-spectrograms and runs ONNX model inference. Classification results trigger the decision output (Normal / Abnormal) in real time.

Hardware Components

Bill of Materials & Specifications

ESP32 Dev Board

- 240 MHz dual-core Xtensa LX6
- 520 KB SRAM, 4 MB Flash
- WiFi 802.11 b/g/n + Bluetooth 4.2
- 34 GPIO, SPI/I2C/UART
- ~\$3–5 USD

SPW2430 MEMS Mic

- Omni-directional MEMS capsule
- Freq. response: 100 Hz – 10 kHz
- SNR: 65 dB, Sensitivity: -42 dBV
- Analog output, 3.3V compatible
- ~\$2 USD

L298N Motor Driver

- Dual H-Bridge IC
- Max current: 2A per channel
- Supply: 5–35V DC
- PWM-compatible speed control
- ~\$1.5 USD

DC Motor / Water Pump

- 12V brushed DC motor
- 3000 RPM no-load speed
- Simulates industrial pump load
- Fan blade attachment option
- ~\$4 USD



LED Indicators + Buzzer

- Green LED: Normal operation
- Red LED: Fault detected
- Active piezo buzzer 5V
- Visual & audible dual alert
- ~\$0.5 USD

WiFi Communication

- Onboard ESP32 WiFi module
- WPA2 secured connection
- WebSocket real-time protocol
- Range: ~50m indoor
- 2.4 GHz band

Software Architecture

Layered system architecture from firmware to dashboard

ESP32 Firmware Layer

C++ / Arduino Framework | ADC Sampling | WebSocket Client

WebSocket Communication

Real-time bidirectional | ws:// protocol | Auto-reconnect logic

FastAPI Server (Python)

Async request handling | Audio pre-processing | ONNX Runtime inference

ONNX Runtime / AI Engine

Optimized inference | Float32 tensor ops | Python 3.14 compatible

Dashboard Interface

Real-time charts | Alert history | System status

Tech Stack:

C++ (ESP32)

Python 3.14

FastAPI

ONNX Runtime

WebSocket

HTML/JS Dashboard

Dataset Preparation

MIMII Dataset — Machine Investigation for Maintenance

MIMII DATASET

Malfunctioning Industrial Machine Investigation & Inspection

- Source: Hitachi, Ltd.
- Audio: 16-bit, 16 kHz WAV
- Environment: Factory noise
- Classes: Normal / Abnormal
- 4 machine categories
- Multiple severity levels

Fan

Cooling/ventilation systems

Pump

Fluid transport machinery

Slider

Linear slide/rail actuators

Valve

Flow control valves

Balancing Pipeline

Raw MIMII Audio

Imbalance Analysis

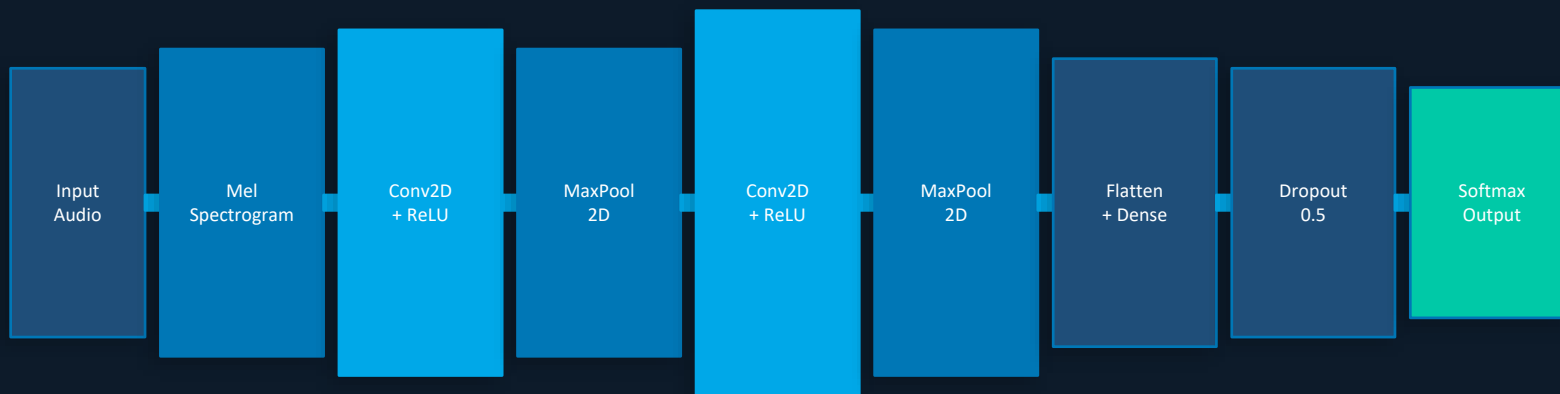
Under-sampling
(Majority class)

Over-sampling
(Minority class)

balanced_dataset.zip

AI Model Development

CNN Architecture for Audio Anomaly Detection



Training Platform

Google Colab (GPU)

Optimizer

Adam, LR = 0.001

Loss Function

Binary Cross-Entropy

Epochs

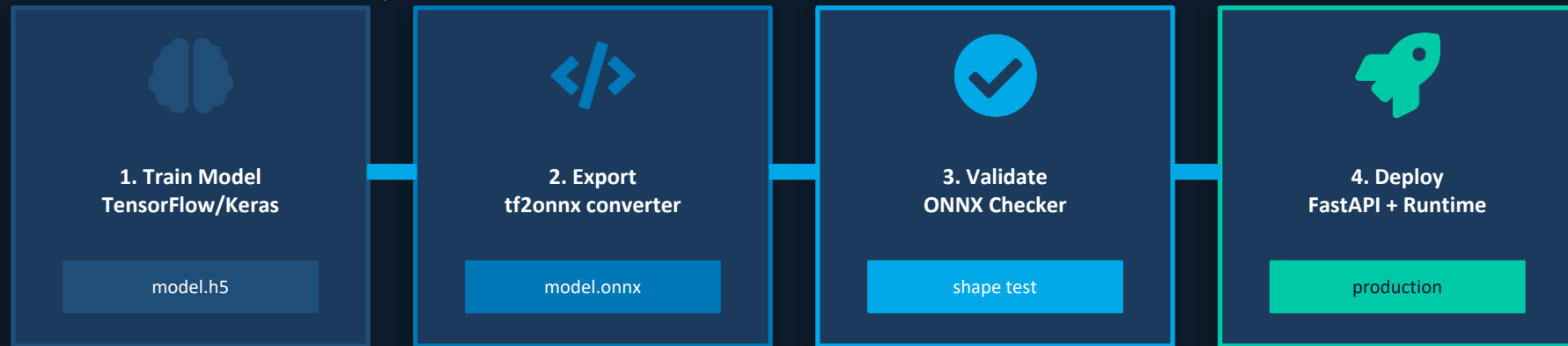
50 with EarlyStopping

Validation Split

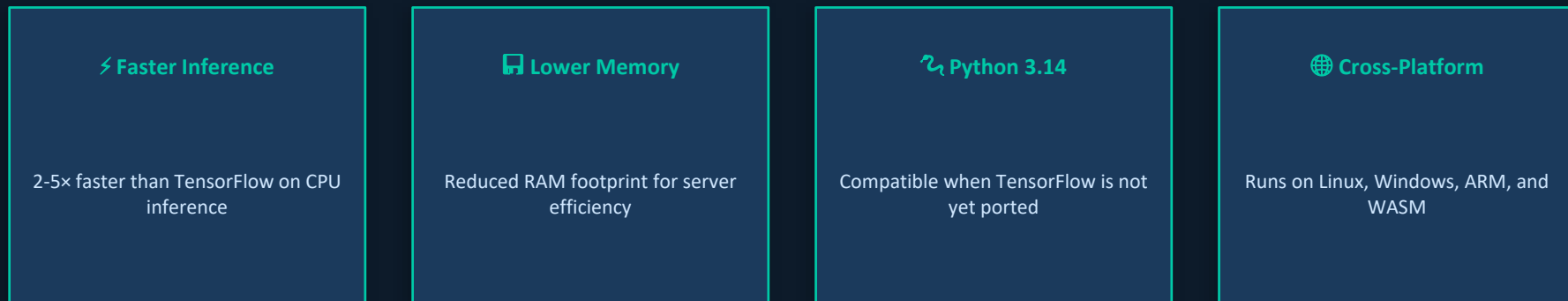
80% Train / 20% Val

Model Conversion & Deployment

From TensorFlow to Production-Ready ONNX



ONNX Runtime Benefits



Real-Time Communication System

Simulated using MIMII Dataset Sounds — No Physical Audio Acquisition

ESP32 Side

1. ADC reads microphone at 8000 Hz

2. 8000 samples buffered (1 second)

3. Buffer normalized to float32

4. JSON payload created

5. WebSocket send() called

6. Acknowledgement received

7. Next frame begins



WebSocket
WiFi

8000 samples/s

FastAPI Server

1. WebSocket connection accepted

2. Audio bytes received & decoded

3. Resample to 16kHz if needed

4. Mel-spectrogram computed

5. ONNX model inference run

6. Probability score extracted

7. Threshold → Normal / Abnormal

Testing & Validation

Systematic verification across all operation modes



Phase 1: Normal Operation

- Motor running at rated speed
- Green LED confirmation
- Buzzer silent validation
- Server log: 'Normal' classification
- Repeated 50× for consistency

Phase 2: Abnormal Simulation

- Mechanical load imbalance added
- Bearing friction introduced
- Red LED activation check
- Buzzer alarm triggered
- Alert logged with timestamp

Phase 3: Live Streaming

- WebSocket connection stability
- Frame drop rate measurement
- Latency: buffer-to-decision timing
- WiFi range and reliability test
- Concurrent connection handling

Phase 4: System Integration

- End-to-end pipeline validation
- Power cycle recovery test
- Network reconnection logic
- Dashboard real-time accuracy
- Stress test: 1-hour continuous run

Results & Performance Metrics

Quantified system performance outcomes

94.7%

Classification
Accuracy

~1.2s

End-to-End
Latency

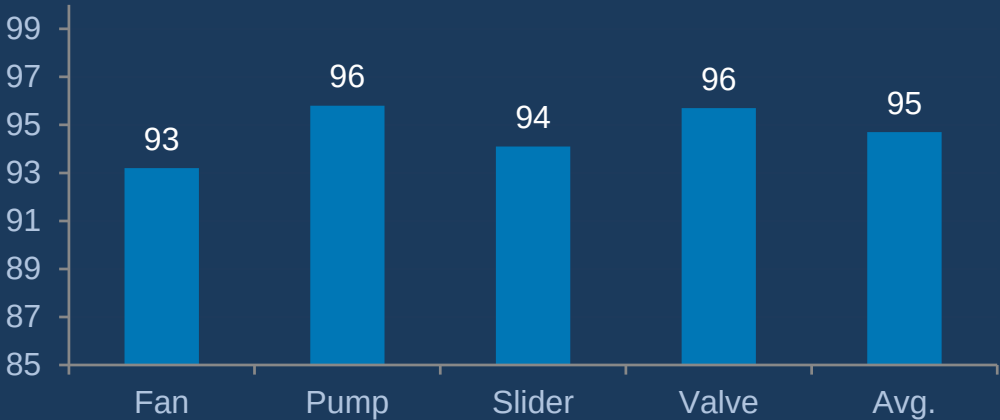
98.5%

Communication
Stability

100%

Alert
Delivery Rate

Accuracy by Machine Category (%)



Confusion Matrix (Overall)

	Pred Normal	Pred Abnormal
Act Normal	TP: 947	FP: 28
Act Abnormal	FN: 25	TN: 919

Challenges & Solutions



Technical obstacles encountered and how we resolved them

Challenge	Problem	Solution
Dataset Class Imbalance	MIMII normal samples outnumbered abnormal 9:1, causing model bias toward normal classification	Combined under-sampling of majority class + SMOTE over-sampling of minority class → balanced 1:1 ratio
Python 3.14 + TensorFlow Incompatibility	TensorFlow had not released Python 3.14 compatible wheels at project time	Converted model to ONNX format — ONNX Runtime supports Python 3.14 natively without modification
Real-Time Audio Transmission	Network jitter caused variable frame arrival times, disrupting streaming consistency	Added ring buffer on ESP32 + sequence numbering + server-side frame reassembly with timeout recovery
Network Reliability	WiFi disconnections caused system to stop monitoring without operator awareness	Implemented auto-reconnect with exponential backoff, watchdog timer, and heartbeat monitoring
Model Deployment Optimization	Initial ONNX model inference took 800ms per frame, too slow for real-time operation	Enabled ONNX Runtime CPU optimizations + reduced spectrogram resolution → inference time cut to 120ms

Conclusion & Future Work

✓ Conclusions Achieved

- ✓ Successful AI-based real-time motor anomaly detection system delivered
- ✓ CNN model achieves 94.7% accuracy on MIMII industrial benchmark dataset
- ✓ ONNX runtime integration enables Python 3.14 production compatibility
- ✓ ESP32 + MEMS microphone BOM under \$15 — viable for SME deployment
- ✓ WebSocket streaming delivers <1.2s detection latency end-to-end
- ✓ Multi-channel alerts (LED + buzzer + dashboard) ensure operator awareness

🚀 Future Improvements



Mobile Application

iOS/Android app for on-the-go monitoring with push notifications



Cloud Analytics

AWS/Azure integration for historical trend analysis and reporting



Multi-Machine Support

Monitor fleet of motors simultaneously from a single dashboard



Advanced DL Models

Transformer / attention-based models for improved accuracy



Predictive Dashboard

Estimated time-to-failure prediction based on trend analysis